EXISTENCE OF A FINITELY GENERATED IDEAL OVER WHICH A FINITELY PRESENTED MODULE HAS A PURE SPREADOUT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AT8 (sources/stacks-flat/pinned-original-source.tex line 7270).

Allowed standard prerequisites (editorial).

- The situation of the reference equation and the notion of a pure spreadout. Same chapter, flat.tex, Equation labelled equation-star (source line 7208), invoked in the statement. Fixes the rings A, B, the localisation S and the module N in which the spreadout is sought.
- Universal injectivity of a map of modules with a free target after dividing out. Same chapter, Lemma labelled lemma-invert-universally-injective (source line 1976), invoked at proof lines 7298-7299. Supplies that the injectivity of multiplication by the elements of S on N'/mN' implies universal injectivity of N'/IN' -> N/IN; the construction of I and of the finite model is inside the counted span.

Source reference. Original equation 0AT6: sources/stacks-flat/pinned-original-source.tex, source lines 7208–7208. Same-chapter pointer into the author environment equation at source line 7208; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 05FR: sources/stacks-flat/pinned-original-source.tex, source lines 1976–1976. Same-chapter pointer into the author environment lemma at source line 1976; invoked at proof line 7299 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In (0AT6) for every $\mathfrak{p} \in \mathrm{Spec}(A)$ there is a finitely generated ideal $I \subset \mathfrak{p}A_{\mathfrak{p}}$ such that over $A_{\mathfrak{p}}/I$ we have a pure spreadout.*

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Proof. We may replace A by $A_{\mathfrak{p}}$. Thus we may assume A is local and $\mathfrak{p}$ is the maximal ideal $\mathfrak{m}$ of A. We may write $N = S^{-1}N'$ for some finitely presented B-module N' by clearing denominators in a presentation of N over $S^{-1}B$. Since $B/\mathfrak{m}B$ is Noetherian, the kernel K of $N'/\mathfrak{m}N' \to N/\mathfrak{m}N$ is finitely generated. Thus we can pick $s \in S$ such that K is annihilated by s. After replacing B by B_s which is allowed as it just means passing to an affine open subscheme of $\mathrm{Spec}(B)$, we find that the elements of S are injective on $N'/\mathfrak{m}N'$. At this point we choose a local subring $A_0 \subset A$ essentially of finite type over $\mathbf{Z}$, a finite type ring map $A_0 \to B_0$ such that $B = A \otimes_{A_0} B_0$, and a finite B_0-module N'_0 such that $N' = B \otimes_{B_0} N'_0 = A \otimes_{A_0} N'_0$. We claim that $I = \mathfrak{m}_{A_0}A$ works. Namely, we have

$$N'/IN' = N'_0/\mathfrak{m}_{A_0}N'_0 \otimes_{\kappa_{A_0}} A/I$$

which is free over A/I. Multiplication by the elements of S is injective after dividing out by the maximal ideal, hence $N'/IN' \to N/IN$ is universally injective for example by Lemma 05FR. $\square$